Simulating RREA with Feedback in an RZ Particle-in-Cell Code

RREA WarpX/AMReX project

Abstract

This note describes a hybrid kinetic/fluid particle-in-cell model of relativistic runaway electron avalanches in a realistic thundercloud field, and the pipeline that turns its output into the quantities instruments record. Kinetic runaway electrons, positrons and photons are coupled to a low-energy fluid whose conductivity discharges the field driving the avalanche, in cylindrical symmetry, on the open-source WarpX and AMReX frameworks. The field solve is Maxwell at finite propagation speed. Optical emission in the photometer bands is driven by the native ion-pair source and transported to the observer through the cloud. This note assumes RREA and relativistic feedback are familiar and particle-in-cell methods are not; the architecture guide remains the reference.

1 What this code is

A thundercloud region above the RREA threshold multiplies runaway electrons; bremsstrahlung photons and pair-produced positrons can travel back upstream and seed fresh avalanches; if each avalanche seeds on average more than one successor, the process becomes self-sustaining. What such an event would look like to an instrument, in shape and timing (absolute amplitudes follow the assumed seed), is what this model exists to predict.

The model solves that whole system at once: everything in Figure 1 is computed in one run, each element reacting to the others every time step.

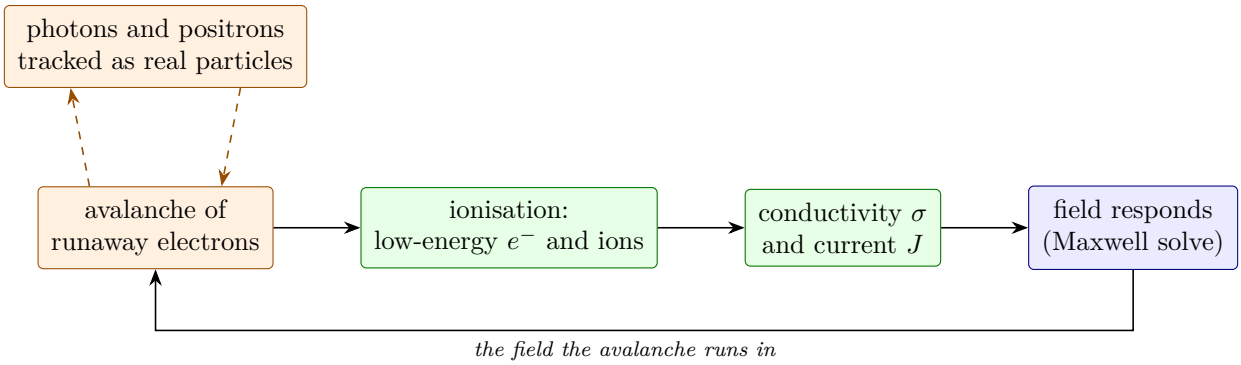


Figure 1: The loop that is closed here. Solid path: the avalanche creates a low-energy plasma, which conducts, which discharges the field, which changes the avalanche the field is driving. Dashed path (left arrow up, right arrow down): photons and positrons are produced by the avalanche and carried as kinetic species, so they can travel back upstream and seed new avalanches.

In one sentence: **this is a hybrid kinetic/fluid, electromagnetic particle-in-cell model in cylindrical symmetry**, built on the open-source WarpX/AMReX stack, with Monte Carlo charged-particle and photon transport from Geant4-anchored interaction tables.

2 How you solve a physics problem with a PIC code

The name is slightly misleading: "particle-in-cell" is less about particles than about the *marriage* of particles and a mesh.

PIC answers two separate impossibilities. Computing every pair interaction is N^2 with N astronomically large; but particles need not see each other individually, only the *field* they collectively produce, so a mesh is introduced as the meeting place. And for TGF, flickering-gamma-ray-flash and glow production N is of order 10^{16} runaway electrons and upward, which no machine can hold or push individually, so the simulation uses weighted samples (Section 4). The loop below addresses the first:

1. **Deposit.** Each particle's charge and current are spread onto the nearby mesh points. The mesh now holds the charge density ρ and the current density J.
2. **Solve.** A field equation is solved *on the mesh*: here, Maxwell's equations at finite propagation speed (Section 6). This is the global calculation, and where the collective physics happens.
3. **Gather.** The field is interpolated from the mesh back to each particle's position.
4. **Push.** Each particle is advanced by the equation of motion under the field it just received.
5. **React.** Particles undergo collisions, ionisation, scattering, creation and loss; for us, Monte Carlo transport in air.

Repeat, one time step at a time. The cost is now roughly linear in particle number plus one field solve, which is what makes the problem tractable at all.

The field and particle updates form a coupled loop. Consistency still requires compatible charge/current deposition, source injection and boundary conditions; PIC alone does not establish it. Here seeds are injected without a current, so the field is that of neutral pairs created in place; the Gauss monitor measures the counter-charge they leave behind.

Two constraints come with the method and set the run cost. The mesh must resolve the structures of interest, since anything smaller is invisible to the field solve; its scale must be compared with the characteristic avalanche length. The outer time step must likewise satisfy the two-dimensional Yee condition $c\Delta t\sqrt{\Delta r^{-2} + \Delta z^{-2}} < 1$, and quantitative claims still require spatial and temporal convergence. Section 4 returns to what that sampling choice costs.

What PIC is good at: self-consistent collective behaviour, and kinetic populations far from thermal equilibrium, both essential here. What it is bad at: representing enormous numbers of low-energy particles individually, and resolving structure below the mesh scale. Both shape Section 6.

3 AMReX and WarpX

AMReX provides the mesh, its distribution across MPI ranks, and the array and communication primitives used by the explicit Maxwell update. WarpX is a PIC code built on AMReX, developed for laser-plasma and accelerator modelling, which is why it may be unfamiliar in this field; it supplies the loop of Section 2 and a well-tested cylindrical (RZ) mode. The RREA physics (the Monte Carlo air interactions, the low-energy fluid, the field solve, the atmosphere profile) is added through an overlay without modifying either dependency's pinned source tree.

A word on why the field solve is ours rather than WarpX's own electromagnetic solver, since that is the obvious question. WarpX's EM loop is bound to its own particle push and current deposition, and this model deliberately replaces the push: the runaway transport owns each particle's trajectory, so WarpX would deposit a path nobody flew. Its RZ material (conductive-medium) update is also absent, and conduction is precisely the physics here. What we need instead is small: axisymmetry with poloidal currents leaves exactly three field components, so the solver is a few hundred lines that read the same face currents the fluid already builds.

4 The macroparticle, and why the population must be managed

A simulated "particle" is not one electron. It is a *macroparticle*: a sample of phase space carrying a weight, the number of physical particles it represents. The production source is a continuous cosmic-ray (PARMA) electron/positron column plus its initial stationary population; the schedule assigns weights so the sampled macros represent the integrated physical fluence.

This is Monte Carlo sampling, not a fudge factor: the macroparticle follows the equations of a single real electron, and the weight enters only when quantities are summed. Weighting is not a convenience either, it is the only reason the problem is computable. Following 10^{16} runaway electrons individually is beyond any foreseeable machine by many orders of magnitude: at a few tens of bytes of state each they would occupy hundreds of petabytes before a single step is taken. Sampling the same population with a finite macroparticle ensemble, each becoming much heavier as multiplication proceeds, is what brings it into reach. The cost is statistical resolution, since with fewer, heavier macroparticles the noise is larger. What weighting does to the physics is a genuine assumption and worth stating as one: for quantities that are simply sums over the population the estimator can be unbiased, and that is the limit of what is guaranteed. Once the sampled population feeds the nonlinear charge–conductivity–field loop, the average of a function is not the function of the average. Increasing the macro count is expected to reduce sampling error, but sufficiency must be established separately for each observable with population-resolution and ensemble ladders.

The complication specific to an avalanche is that the population *multiplies*, so macroparticle count would grow until memory ran out. When the population reaches its control band the code performs a **charge-exact stochastic thinning**. It adjusts weights along the null space of the four-cell CIC deposit, eliminating particles while leaving every deposited cell charge unchanged in that realization. Each particle's expected weight is preserved, so thinning costs particle/spectral statistics without injecting charge noise into the Maxwell field.

5 Why cylindrical (RZ), and what it costs

The simulation runs in cylindrical coordinates (r, z) with azimuthal symmetry: every quantity is independent of azimuth. The mesh is therefore two-dimensional, while the physics (volumes, densities, the field solve) stays fully three-dimensional, since each cell is a ring of revolution.

What this buys is decisive. Resolving the same radial extent with a full three-dimensional mesh adds the missing azimuthal cells and makes kinetic transport impractical. RZ replaces those cells by physically weighted rings, making the production calculation tractable.

What it forbids is real: no azimuthal structure, no tilted or off-axis geometry, no filamentary channels that break symmetry. The model describes an axisymmetric avalanche column.

Axisymmetry also makes the exact axis a radiation null, which is why the on-axis diagnostic of Section 8 is a longitudinal field, not radio.

6 The hybrid: kinetic runaways, fluid low-energy plasma

Section 2 noted that PIC is bad at carrying enormous numbers of low-energy particles. An RREA produces many more low-energy carriers than runaways. Representing those as macroparticles would swamp the run, and would be wasteful, because they matter almost entirely through the *conductivity* they provide.

So the model splits the plasma:

Kinetic species	runaway electrons, positrons, photons, each a macroparticle with position, momentum and weight. Transport follows them well below 1 MeV; the 1 MeV threshold in the figures (which label these "energetic electrons") is a display cut, not a physics cutoff
Fluid	low-energy electrons, positive and negative ions, as densities on the mesh, with mobilities and attachment, contributing conductivity σ

Photons and positrons are kinetic *because feedback requires it*: their whole significance is that individual quanta travel long distances upstream and deposit a new seed elsewhere, which is a transport problem a fluid cannot represent. Photons interact through Compton scattering, photoelectric absorption and pair production, all carried explicitly. Which one dominates depends on photon energy and on the question being asked, so the code privileges none of them.

The transport is table-driven: cross sections and secondary-particle distributions are precomputed into a versioned table set anchored to Geant4 and loaded at run time. One table set serves every altitude because the tables are built at a reference density and scaled by the local air density.

The field solve. This is where the self-limiting behaviour lives: as σ rises, the current it carries drains the field driving the avalanche, and the avalanche self-quenches. The field equation is Maxwell's at finite propagation speed. Because the avalanche is axisymmetric and every current is

poloidal (J_r, J_z), the complete field content is the three-component TM set (E_r, E_z, B_θ), advanced on a staggered Yee mesh by an explicit leapfrog:

- field changes propagate at c, so the collapse timing is computed rather than assumed;
- conduction enters through the same face currents that move the fluid carriers, so the screening current in the field equation and the continuity current are one object; both subcycle together when conductivity shortens the screening time, while the outer step stays fixed.

Three practical points. The solver advances the *scattered* field over the static thundercloud background. The outer boundaries are first-order Silver–Müller absorbers, which reduce but do not eliminate reflections. Gauss's law is monitored, not enforced; see `docs/RREA_RADIO_LIMITATIONS.md`.

A standing rule about profiles. Scientific production and campaign captures use the altitude-dependent field and NRLMSISE-00 density profiles, never a constant-atmosphere substitute. Avalanche length and multiplication vary strongly with altitude, so a constant-density, constant-field calculation answers a different question.

7 How to read a production capture

Every capture records its resolved source, geometry, profiles, population policy and numerical controls. Read population, current, field and observer diagnostics together, and check conservation, convergence and the stated model scope.

The standard video places six r–z maps beside one altitude profile axis of the radial-mean and on-axis field. The maps show energetic electrons, positive ions, photons, positrons, absolute field magnitude and field change relative to the ambient peak. Fixed colour/profile scales make the temporal evolution directly comparable.

8 Radio and field-change observables

The Maxwell TM solver records the scattered (E_r, E_z, B_θ) field on a closed surface inside its absorbing boundary. `scripts/compute_rrea_em_radio.py` evaluates that surface at an observer on the symmetry axis. The result is useful, but it is *not radio*: for an axisymmetric source, $E_r = B_\theta = 0$ on the axis and the outward Poynting flux there is exactly zero. The surviving E_z is a longitudinal Coulomb/induction field suitable for a nadir field-mill diagnostic.

For a nonzero horizontal observer offset the same script evaluates the full vector Love-equivalence surface integral by azimuthal quadrature. That is the direct finite-source free-space observable; ground, ionosphere, terrain and calibrated antenna gain are outside scope.

The standard multiband figure combines above-12-km electron/photon populations, optical estimates, and these reduced Huygens observer fields. When observer reductions are absent, it uses explicitly labelled current-moment/line-source approximations. Those approximations collapse the radial source distribution; they are not interchangeable with the full surface integral.

9 Optical: where the 337.1 and 777.4 nm light comes from

The optical pipeline uses measured air fluorescence. Ionizing cascades produce excited molecules, whose light competes with collisional quenching.

The degradation term uses the zero-field laboratory air-fluorescence yield as its baseline: the Rosado et al. world average with AIRFLY's measured pressure and temperature quenching. The engine records the cumulative ion-pair count directly, so the optical baseline uses that source multiplied by the measured energy per pair. The transport energy-loss ledger is printed alongside as a consistency diagnostic, not treated as an independent deposition estimator.

Field-driven excitation is assessed separately. Selected Phelps/LXCat cross sections from the MCAIR electron Monte Carlo project's data feed an offline BOLSIG calculation. The low-electron exposure, sampled maximum field and independent quenching limits give a sampled model envelope for omitted swarm emission. The plotted 337-nm curve remains deposition fluorescence. A missing assessment is labelled *unassessed*; a numerically truncated tail is not declared physically zero. Native ionization-source centroids also bound the fluorescence-yield error from using one altitude, although they do not recover the source width or radial distribution needed for a precise propagated pulse. Field-induced changes in the prompt degradation spectrum remain a separate model uncertainty; the small relaxed-swarm envelope does not bound them.

777.4 nm must not be read as a TGF prediction. There is no leader in this simulation, and real instruments see strong 777.4 nm emission from hot leader channels. What the model produces is a cold-discharge estimate. (The line itself is atomic oxygen, produced here by dissociative excitation of O_2 and heavily quenched at these altitudes; it is not a nitrogen band.)

Getting the light to the observer requires atmospheric transport. Photons are carried through the thundercloud to the observer by Monte Carlo radiative transfer (CloudScat). **The assumed cloud top dominates this result and must always be quoted** together with the capture's measured emission-weighted source altitude. Their separation sets the optical depth, delay and pulse width, so the delay is a property of the assumed cloud; that the scattered fluence can *exceed* the free-space value is genuinely physical, since scattering redirects into the line of sight light that would otherwise have missed the observer.

10 What is checked, and what is not

What is checked. The focused tests pin governing equations, dimensions, conservation, causality, limiting cases and production-path wiring. Transport samplers are compared with analytic distributions and independent Geant4 holdouts not used to fit the runtime tables. The low-energy closure checks primary-source dry-air rows from Milloy, Shimamori–Hatano, Hegerberg–Reid, Morrow–Lowke, Dutton and de Urquijo, including the thermal limit, tensor diffusion and the exclusion of humid attachment values.

What is not checked. No observed TGF lightcurve, calibrated photometer trace or measured sferic is treated as ground truth. The Coleman–Dwyer avalanche-length prescription provides an independent pure-RREA comparison over the real altitude profile and both 276/284-kV/m threshold conventions, but feedback-enabled results need not reproduce it. Confidence in the unexplored full system comes from conservation, convergence, causality, parameter sensitivity and agreement in the

sub-regimes that do have external anchors.

What the model does not contain, stated plainly:

- No leader and no streamer-scale channel, which bounds the optical interpretation above.
- No geomagnetic field, and no azimuthal structure (Section 5).
- No coherent Rayleigh scattering in the kinetic gamma-ray transport, a deliberate exclusion that bounds near-cut angular transport and feedback seeding; the optical CloudScat transfer is separate and does include it.
- Cold-cloud and humidity corrections to the room-temperature dry-air electron closure, and species-resolved ion chemistry; the dry-air closure table stops at 60 Td, since the fluid has no impact-ionisation channel.
- Charged transport is first order in the outer step; quantitative claims need timestep and population ladders (Section 4).
- A leading zero in a sparsely sampled population lightcurve cannot by itself establish a physical arrival front; the curves are populations inside a region, never a flux through it.

The full architecture document (`docs/RREA_WARPX_AMREX_ARCHITECTURE_GUIDE.tex`) carries the detail behind every choice summarised here, including the ordering contract with WarpX, the table schema, the field-solve details, and the complete observable derivations.